W13)

Prove each of the following statements

a) $(L^*)^* = L^*$ b) $L^* \cdot L^* = L^*$ c) $(L^+)^+ = L^+$

Answer: $(L^*)^* = \bigcup_{n=0}^{\infty} (L^*)^n$

- $\forall w \in (L^*)^* \Rightarrow \exists n \in \mathbb{N}$ with $w \in (L^*)^n = \underbrace{L^* \cdot L^* \cdots L^*}_{n \text{ times}}$

$\Rightarrow \exists w_1, w_2, \ldots, w_n \in L^*$ with $w = w_1 \cdot w_2 \cdots w_n$

$w_1 \in L^* \Rightarrow \exists n_1 \in \mathbb{N}$ with $w_1 \in L^{n_1}$, then $w_1 = w_{11} w_{12} \cdots w_{1n_1}$ with $w_{1i} \in L \ \forall i \leq n_1$

the same for $w_2, \ldots, w_n$

then $w = (w_{11} \cdot w_{12} \cdot \cdots \cdot w_{1n_1}) \cdot (w_{21} \cdot w_{22} \cdot \cdots w_{2n_2}) \cdots$

$\cdot (w_{n1} \cdot w_{n2} \cdot w_{n3} \cdot \cdots \cdot w_{nn_n})$

with $w_{ij} \in L \ \forall i$ and $\forall j$ and $n_1 + n_2 + \cdots + n_n \in \mathbb{N}$

$\Rightarrow w \in L^{n_n} \subseteq L^*$ $\boxed{(L^*)^* \subseteq L^*}$ ①

- $(L^*)^* = \bigcup_{n=0}^{\infty} (L^*)^n = \{\varepsilon\} \cup L^* \cup (L^*)^2 \cup \cdots \cup (L^*)^n \cup \cdots$

$\Rightarrow \boxed{L^* \subseteq (L^*)^*}$ ②

- ① and ② $\Rightarrow (L^*)^* = L^*$

b) $L^* \cdot L^* = L^*$

<u>Answer:</u>

$\forall w \in L^* \cdot L^* \Rightarrow \exists w_1 \in L^*, \exists w_2 \in L^*$ with $w = w_1 w_2$

$w_1 \in L^* \Rightarrow \exists n_1 \in \mathbb{N}$ with $w_1 \in L^{n_1}$, then $w_1 = w_{11} w_{12} \cdots w_{1n_1}$ with $w_{1i} \in L \ \forall i \leq n_1$

$w_2 \in L^* \Rightarrow \exists n_2 \in \mathbb{N}$ with $w_2 \in L^{n_2}$, then $w_2 = w_{21} \cdot w_{22} \cdots w_{2n_2}$ with $w_{2i} \in L, \forall i \leq n_2$

$w = w_1 w_2 = w_{11} w_{12} \cdots w_{1n_1} w_{21} \cdot w_{22} \cdots w_{2n_2} \in L^{n_1+n_2}$

$\Rightarrow w \in L^* = \bigcup_{n=0}^{\infty} L^n$ and $\boxed{L^* L^* \subseteq L^*}$ ①

o) $L^* L^* = L^* \cdot \left(\bigcup_{n=0}^{\infty} L^n\right) = L^* \left(L^0 \cup L^1 \cup L^2 \cdots \cup L^n \cdots\right)$

$= L^* \left(\{\varepsilon\} \cup L \cup L^2 \cdots \cup L^n \cdots\right)$

$= L^* \cup L^+ \cup L^* L^2 \cup \cdots \cup L^* L^n \cdots$

and $\boxed{L^* \subseteq L^* \cdot L^*}$ ②

o) ① and ② $\Rightarrow L^* L^* = L^*$

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c) $(L^+)^+ = L^+$

$L^+ = L \cdot L^* = \bigcup_{n=1}^{\infty} L^n \qquad (L^+)^+ = \bigcup_{n=1}^{\infty} (L^+)^n$

•) $\forall w \in (L^+)^+ \Rightarrow \exists n \in \mathbb{N}^*$ with $w \in (L^+)^n = \underbrace{L^+ \cdot L^+ \cdots L^+}_{n \text{ times}}$

$\Rightarrow \exists w_1, w_2, \ldots, w_n \in L^+$ with $w = w_1 \cdot w_2 \cdots w_n$

$w_1 \in L^+ \Rightarrow \exists n_1 \in \mathbb{N}^*$ with $w_1 \in L^{n_1}$, then $w_1 = w_{11} w_{12} \cdots w_{1n_1}$

with $w_{1i} \in L, \forall i \in [1, n_1]$

the same for $w_2, \ldots w_n$

$w = (w_{11} \cdot w_{12} \cdot \ldots \cdot w_{1n_1}) \cdot (w_{21} \cdot w_{22} \cdot \ldots \cdot w_{2n_2})$

$\cdot \ldots \cdot (w_{n1} w_{n2} w_{n3} \ldots w_{nn_n})$

with $w_{ij} \in L \ \forall i$, and $\forall j$

then $w \in L^{(n_1 + n_2 + \ldots + n_n)}$ with $n_1 + n_2 + \ldots + n_n \in \mathbb{N}^*$

$\Rightarrow w \in L^+ \quad \boxed{(L^+)^+ \subseteq L^+}$ ①

•) $(L^+)^+ = \bigcup_{n=1}^{\infty} (L^+)^n = L^+ \cup (L^+)^2 \cup (L^+)^3 \cup \ldots \cup (L^+)^n \ldots$

then $\boxed{L^+ \subseteq (L^+)^+}$ ②

① and ② $\Rightarrow (L^+)^+ = L^+$

W#4)

Give the complement of the following languages:

a) $r = (a+b)^* b$

$\bar{r} = \varepsilon + (a+b)^* a$

Proof:

$r + \bar{r} = \varepsilon + (a+b)^* b + (a+b)^* a$

$= \varepsilon + (a+b)^*(b+a) = \varepsilon + (a+b)^+$

$= (a+b)^*$

b) $r = ((a+b)(a+b))^*$

$\bar{r} = ((a+b)(a+b))^*(a+b)$

Proof:

•) $r + \bar{r} \subseteq (a+b)^*$

•) $\forall w \in (a+b)^*$

Case 1: $|w|$ is even then $w \in ((a+b)(a+b))^*$

Proof by induction using the string length

for $|w| = 0 \Rightarrow w = \varepsilon \in ((a+b)(a+b))^0$

for $|w| = 2 \Rightarrow w = ab$ or $w = ba$ or $w = aa$ or $w = bb$

$ab \in ((a+b)(a+b))^1$ and $ba \in ((a+b)(a+b))^1$

Suppose that w of length $2n$ belongs to $((a+b)(b+a))^n$

take ~~no~~ w a string of length $2(n+1)$

say $w = w_1 ab$ with $|w_1| = 2n$

then $w_1 ab \in ((a+b)(b+a))^n ((a+b)(a+b)) = ((a+b)(a+b))^{n+1}$

the same in case $|W|$ is odd

$(\alpha+\beta)^* = (\alpha^*+\beta^*)^* = (\alpha^*\beta)^*\alpha^* = \alpha^*(\beta\alpha^*)^* = (\alpha^*\beta^*)^*$

$\alpha(\beta\alpha)^* = (\alpha\beta)^*\alpha$ $\qquad\qquad$ $\varepsilon + LL^* = L^*$

WQ5)

Simplify the following regular expression;

a) $r = (\varepsilon + a^* + b^* + a + b)^* = (a^*+b^*)^* = (a+b)^*$

b) $r = a(a+b)^*b + (ab)^* + (ba)^* = a(a+b)^*b + (ba)^*$

c) $r = \phi + \varepsilon + a + (a+b)^* + b(a+b)^*$
$= \varepsilon + (a+b)(a+b)^* = (a+b)^*$

d) $r = a^* + (a^*b^*)^* b(a+b)^* = a^* + a^*(ba^*)^* b(a+b)^*$
$= a^* + a^* b(a^*b)^*(a+b)^*$
$= a^* + a^* b(a^*b)^* a^*(ba^*)^*$
$= a^* + a^* b(a^*b)^*(a^*b)^* a^*$
$= a^* + a^* b(a^*b)^* a^*$
$= a^* + a^*(ba^*)^* b a^*$
$= a^*(\varepsilon + (ba^*)^* ba^*)$
$= a^*(ba^*)^* = (a+b)^*$

e) $((0+1)(0+1)^*)^*$

$((0+1)(0+1)^*)^* = ((0+1)^+)^* = \varepsilon + ((0+1)^+)^+$
$= \varepsilon + (0+1)^+ = (0+1)^*$

f) $(1+01)^*(0+1^*)$
$= (1+01)^*0 + (1+01)^*1^* = (1+01)^*0 + (1^*(01))^*1^*1^*$
$= (1+01)^*0 + (1^*(01))^*1^* = (1+01)^*0 + (1+01)^*$
$= (1+01)^*(0+\varepsilon)$

g) $(b + (\varepsilon+a)(\varepsilon+a)^* b) + (b + (\varepsilon+a)(\varepsilon+a)^* b)$

$\Rightarrow ((\varepsilon+b) + a(\varepsilon+a)^* b)^* ((\varepsilon+b) + a(\varepsilon+a)^* b)$

$= (\varepsilon + (\varepsilon+a)(\varepsilon+a)^*) b + (\varepsilon + (\varepsilon+a)(\varepsilon+a)^*) b$

$\Rightarrow ((\varepsilon+b) + a(\varepsilon+a)^* b)^* ((\varepsilon+b) + a(\varepsilon+a)^* b)$

$= (\varepsilon+a)^* b [\varepsilon + ((\varepsilon+b) + a(\varepsilon+a)^* b)^* ((\varepsilon+b) + a(\varepsilon+a)^* b)]$

$= a^* b ((\varepsilon+b) + a(\varepsilon+a)^* b)^* = a^* b (\varepsilon + b + aa^* b)^*$

$= a^* b (\varepsilon + (\varepsilon + aa^*) b)^* = a^* b (\varepsilon + a^* b)^* = (a^* b)(a^* b)^*$

$= (a^* b)^+$

$\Rightarrow L = \{ w \mid w \text{ ends with letter } b \}$

l) $(b + aa^* b) + (b + aa^* b)(a + ba^* b)^* (a + ba^* b)$

$= (a^* b) + (a^* b)(a + ba^* b)^* (a + ba^* b)$

$= (a^* b)[\varepsilon + (a + ba^* b)^* (a + ba^* b)]$

$= a^* b (a + ba^* b)^*$

$= (a^* b)[a^* (ba^* b)^*]^*$ with $(a+b)^* = (a^* b^*)^*$

with $(a^* b^*)^* = a^* (ba^*)^*$

$= (a^* b) a^* ((ba^* b) a^*)^* = (a^* b) a^* (ba^* ba^*)^*$